

Quantitative Sales Forecasting Practice Case Study

IAL Business practice using three period moving averages variation and evaluation

Extract A VoltRide Ltd

VoltRide Ltd is a UK retailer of electric bicycles, replacement batteries and accessories. Its customers include commuters, leisure cyclists and delivery riders. The business sells through its website and two city-centre stores.

Demand for electric bicycles has grown as more customers seek lower-cost and environmentally friendly travel. However, e-bikes are relatively expensive purchases. Demand can fall when interest rates or inflation reduce consumers' disposable income.

VoltRide's sales increased strongly until 2023. Sales fell in 2024 after a low-price online competitor entered the market and higher interest rates reduced consumer spending. Sales recovered in 2025 after new cycle lanes were introduced in several cities and VoltRide launched a compact commuter e-bike.

The directors are considering leasing a larger warehouse and increasing delivery capacity in 2026. The lease would cost £0.8m each year and would require VoltRide to hold more e-bike stock. A forecast could help managers decide whether the extra capacity is justified.

Extract B Sales revenue data

The table shows VoltRide's annual sales revenue. Complete the moving average and variation columns as part of your preparation for the questions. Variation = actual sales – three period moving average.

Year	Sales revenue £m	Three period moving average £m	Variation £m
2020	8.0		
2021	9.6		
2022	11.2		
2023	13.5		
2024	12.1		
2025	14.4		

A trend line based on the moving averages gives an extrapolated sales forecast of £15.10m for 2026.

Questions

1. Using the data in Extract B, calculate the three period moving average for 2023 and the average variation from trend for 2021 to 2024. You are advised to show your working. **(4)**

1b. Using the data in Extract B, calculate the average variation from trend for 2021 to 2024. You are advised to show your working. **(4)**

2. Explain one conclusion VoltRide could make from the variation figures in Extract B. **(4)**

Use the case evidence and the data. Build a balanced response: explain benefits, analyse limitations and reach a supported judgement.

3. Assess the usefulness of quantitative sales forecasting for VoltRide Ltd. **(12)**

4. Evaluate whether VoltRide should use its adjusted 2026 forecast to lease a larger warehouse and increase delivery capacity. **(12)**

Planning prompt: identify the trend; use at least one variation figure; link the forecast to a decision; explain a limitation; reach a judgement that depends on the stability of the market and the quality of the data.

Plan before you write

Paragraph	Your point	Case evidence or data
Benefit		
Further benefit		
Limitation		
Judgement		